

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459938.78996 +/- 0.00098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1322 +/- 0.0072
Transit depth (Rp/Rs)^2	1.75 +/- 0.19 [%]
Orbital Inclination (inc)	81.57 +/- 1.79
Airmass coefficient 1 (a1)	1.1335 +/- 0.0059
Airmass coefficient 2 (a2)	-0.12 +/- 0.02
Scatter in the residuals of the lightcurve fit is	0.52 %
Best Comparison Star	None
Optimal Aperture	4.67
Optimal Annulus	8.88
Transit Duration (day)	0.1221 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1322 $\pm$ 0.0072 vs 0.1178 (deviation 2.0 σ)
Duration fit vs published	0.1221 d vs 0.125 d (deviation 0.66 σ)
Mid-time O-C	1.8 min
Depth SNR	18.4
Residual scatter	0.52 %

Light curve

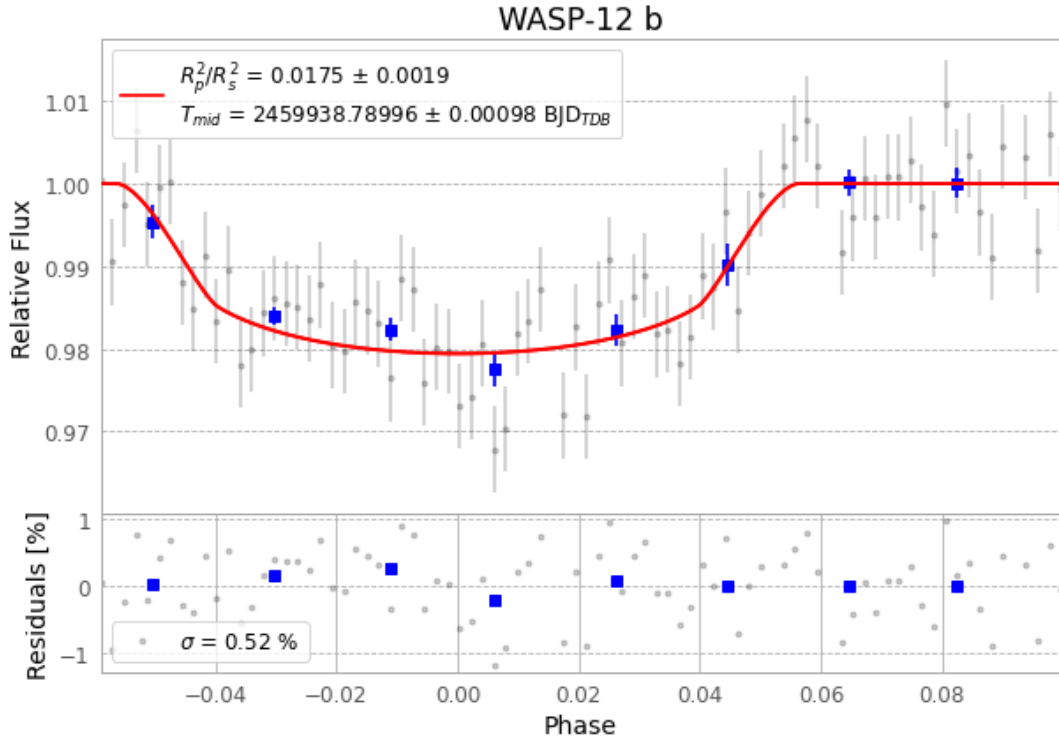


Figure 1 — FinalLightCurve_WASP-12 b_2022-12-24.png (SHA-256 45de81e3ab05f653...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [417, 132], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 70d39322b111a3c1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

84 FITS frames (55 MB), WASP-12221225051515.FITS → WASP-12221225092415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-221225.log (09a68f78e255...), FinalLightCurve_WASP-12 b_2022-12-24.png (45de81e3ab05...), FinalLightCurve_WASP-12 b_2022-12-24.csv (b472e7dc0927...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 09:32Z.